

Robustness investigation: results

A mizer reimplementation of Law & Plank (2023), Fish and Fisheries

What came out of the plan in [robustness.md](#). Phases 1, 2, 4 and 5 are complete, as are Phase 3 items 8, 8b, 10 and 11. Everything below is measured, not argued, and where a measurement contradicts something I wrote earlier in the plan or in an earlier phase, the correction is stated rather than quietly applied.

What survives. The paper's qualitative conclusion is robust. Replicated over thirteen independently assembled ecosystems, the ranking $BH_P > \text{fixed } F > BH_{P/B}$ holds **13 times out of 13**, and it does not depend on how yields are matched: each rule's whole yield-biodiversity frontier was computed, and they never cross. The effect size varies fourfold across ecosystems (median 2.4×, range 1.3–5.2×), so single-ecosystem figures should be read as illustrative.

What it depends on. Three of the paper's choices carry the result. Imposing a stock-recruitment relationship on the published ecosystem, holding the community exactly fixed, shrinks BH_P 's advantage from 3.4× to 1.0×: the density dependence the harvest rule supplies is largely redundant with the density dependence a real population already has, and the model has none of the latter by deliberate design. And the plankton cannot be made contestable without destroying the large-bodied community the paper harvests at all — richness falls from 15 species to 5 and nothing reaches the 400 g entry size — so the inexhaustible resource is doing more work than merely suppressing the recruitment brake. The third is the 400 g entry size: the conclusion holds at every entry size tested, but the advantage runs from 1.2× at 100 g to 3.7× at 800 g, because raising the entry size is what creates the recruitment overfishing that the feedback then prevents.

What is not the mechanism the paper describes. F proportional to *biomass* matches or beats F proportional to production in all thirteen ecosystems, so production is a proxy for biomass rather than the active ingredient — which matters, since biomass is far easier to estimate. Within the family $F_i \propto B_i^\theta$ the benefit saturates by $\theta = 0.5$: only the sign and rough magnitude of the density dependence matter.

What is the mechanism. The adaptive feedback, and it took thirteen ecosystems to establish that — on one it looked as though the initial allocation did most of the work. Freezing BH_P 's year-0 allocation recovers a median of only 38% of the benefit, with a range from −4.7 to +0.9, and is often worse than fishing everything at the same rate. The feedback is what stops the allocation going stale.

What would make it work better. Measuring P and B over the whole life cycle rather than only the harvested range improves both balanced-harvesting rules, and improves $BH_{P/B}$ threefold — so part of that rule's poor showing in the paper comes from the restricted measurement window rather than from the idea of a constant exploitation ratio.

How it would fare in practice. The feedback tolerates infrequent updating remarkably well — every 5, 10, even 25 years is as good as continuous, because the ecosystem’s own timescale is decades. It does not tolerate bad estimates: with observation error of log-scale s.d. 1.0, BH_P becomes *worse* than fixed F on average. It protects against structural uncertainty, not observational uncertainty.

Reproduce with `run_robustness.R` (phases 1, 2), `run_resource_sweep.R` (item 8b), `run_recruitment.R` (item 8), `run_replication.R` (phase 5), `run_implementation.R` (phase 4) and `run_fishery_design.R` (items 10, 11).

The experiments, stated precisely

Notation follows the paper: species $i = 1, \dots, n$, body mass w , number density $N_i(w, t)$ per unit mass per m^2 . The model grid is $w_1 < \dots < w_J$ with $w_1 = 1$ mg and bin widths Δw_j , and $g_{ij}(t)$ is mizer’s `e_growth`, the somatic growth rate dw/dt in $g\ yr^{-1}$ — equal to $\epsilon_i(w) \tilde{g}_i(w, t)$ in the paper’s Appendix A notation. Every experiment uses $T = 50$ years of fishing. The model itself, equation by equation, is in [the overview](#).

The ecosystems are named after the seed they were assembled from, by the Appendix B procedure described in [the overview](#): `eco1` is the 15-species assemblage behind the paper’s Figs 2-4; `eco_r1` - `eco_r3` are three more with the search rate randomised across species; and `eco301` - `eco312` are twelve further ones assembled under exactly the `eco1` protocol, used for the replication in Phase 5. None of them is the paper’s own assemblage, which is not recoverable from what the paper reports.

The harvest rules

The fishery is knife-edged at a shared entry mass w_f , taken as the first grid point at or above it,

$$j_f = \min \{ j : w_j \geq w_f \}.$$

All rules have the same form,

$$F_i(w_j, t) = F_i(t) \mathbf{1}[j \geq j_f], \quad F_i(t) = c z'_i g_i(t),$$

differing only in the per-species allocation $g_i(t)$:

$$g_i = \begin{cases} 1 & \text{fixed, Eq. (2.7)} \\ P_i(t) & BH_P, \text{ Eq. (2.8)} \\ P_i(t) / B_i(t) & BH_{P/B}, \text{ Eq. (2.9)} \\ B_i(t) & BH_B \\ B_i(t)^\theta & \text{power } (\theta) \\ h_i & \text{frozen } (h), \text{ constant in } t \end{cases}$$

Negative or non-finite F_i are set to zero. z'_i is the per-species fishing intensity factor: $z'_i = 1$ for `eco1`, and $z'_i \sim U(0.5, 1.5)$ (seed 99) for `eco_r1`-3, held fixed across rules within an

ecosystem, as in the paper's Fig. 6.

The frozen controls take $h_i = g_i^X(0)$, the year-0 allocation of rule X , and hold it constant for the whole 50 years. Comparing frozen ($g^{\text{BHP}}(0)$) with BHP separates *allocating* effort across species from *updating* that allocation as the ecosystem changes.

B_i and P_i , as discretised

Over the harvested range $[w_f, w_{\max, i}]$, with J the top grid index:

$$B_i(t) = \sum_{j \geq j_f} w_j N_{ij}(t) \Delta w_j$$

$$P_i(t) = \underbrace{\sum_{j \geq j_f} g_{ij}(t) N_{ij}(t) \Delta w_j}_{\text{production inside the range}} + \underbrace{w_{j_f} g_{ij_f}(t) N_{ij_f}(t)}_{\text{flux in at } w_f} - \underbrace{w_J g_{iJ}(t) N_{iJ}(t)}_{\text{flux out at the top}}$$

The boundary terms are the ones integration by parts demands; the flux out vanishes because $\epsilon_i \rightarrow 0$ at $w_{\max, i}$. Yield is $Y_i(t) = F_i(t) B_i(t)$, Eq. (2.6), which is exact because B_i uses the same j_f as the selectivity.

$B_i^{\text{tot}}(t)$ denotes the same sum over the *whole* grid, $j \geq 1$. The biodiversity measures below use B^{tot} ; the yields use B_i .

Anchoring the intensity constant

Each rule's constant is set so that the *initial* total yield equals that of fixed fishing at a reference rate F_{ref} . Since $Y(0) = \sum_i F_i(0) B_i(0) = c \sum_i z'_i g_i(0) B_i(0)$,

$$c_X^0 = F_{\text{ref}} \frac{\sum_i B_i(0)}{\sum_i z'_i g_i^X(0) B_i(0)},$$

which is closed-form and needs no simulation. $F_{\text{ref}} = 0.1 \text{ yr}^{-1}$ for eco1 and the plankton variants, and 0.2 yr^{-1} for eco_r1-3, matching the paper's doubled baseline for its Fig. 6. The sweep then runs $c = m c_X^0$ for

$$m \in \{0.1, 0.25, 0.5, 1, 2, 4, 8\} \quad (\text{Phases 1, 2 and the recruitment sweep}),$$

$$m \in \{0.25, 0.5, 1, 2, 4\} \quad (\text{plankton variants}).$$

Outcome measures

The unfished control is the same params object with $F_i \equiv 0$, started from the same state and run with the same Δt and T , so that numerical and drift effects cancel. Writing

$$R_i = \frac{B_i^{\text{tot}}(T)}{B_i^{\text{tot, ctrl}}(T)},$$

the measures are

$$\begin{aligned}
\text{worst species} &= \min_i R_i, \\
\text{below 10\%} &= \#\{i: R_i < 0.1\}, \\
\text{RMS log} &= \sqrt{\frac{1}{n} \sum_i (\log \max(R_i, 10^{-12}))^2}, \\
\text{terminal yield} &= \sum_i F_i(T) B_i(T), \\
\text{cumulative yield} &= \int_0^T \sum_i Y_i(t) dt \quad (\text{trapezoid on the saved times}).
\end{aligned}$$

The 10^{-12} floor only binds for species already extinct by any standard.

Comparing rules at a common yield

Each rule gives a set of (terminal yield, metric) points along its frontier. A metric is read at a yield none of the runs hit exactly by interpolating linearly in $\log(\text{yield})$; log-log if the metric is strictly positive along the frontier, otherwise linear in the metric (which is what the integer “below 10%” count uses). The reference yield is always the fixed- F run at $m = 1$ in that ecosystem.

Plankton variants

In mizer’s per-mass densities the paper’s Eqs (A.12)–(A.14) read

$$r(w) = r_0 w^{-\rho}, \quad a(w) = \frac{a_0}{w} \left(\frac{w}{w_a}\right)^{1-\lambda}, \quad I(w) = \frac{I_0}{w} \left(\frac{w}{w_I}\right)^{1-\lambda},$$

with $\rho = 0.15$, $\lambda = 2$, $w_a = w_I = 1$ mg. Total primary production is $\int r(w) w N_{pp}(w) dw$, which for an undepleted spectrum is proportional to $r_0 a_0$. The variants therefore hold

$$r_0 a_0 = 2 \times 10^4 \quad (\text{the published pair is } 10, 2000),$$

so that they differ in how *contested* the plankton is, not in how much of it there is. Variants used, as (r_0, a_0, I_0) : $(10, 2 \times 10^3, 2 \times 10^3)$ published, $(3, 6.67 \times 10^3, 2 \times 10^3)$, $(1, 2 \times 10^4, 2 \times 10^3)$, and in the single-species screen also $(0.5, 4 \times 10^4, 2 \times 10^3)$ and $(0.1, 2 \times 10^5, 0)$.

Resource level

Writing $L = N_{pp}/a$ and $\iota = I/a$, the steady state of Eq. (A.11) at grazing mortality d is

$$\begin{aligned}
\iota + r L (1 - L) - d L &= 0 \quad \Rightarrow \quad r L^2 - (r - d) L - \iota = 0, \\
L &= \frac{(r - d) + \sqrt{(r - d)^2 + 4r\iota}}{2r}.
\end{aligned}$$

This is the same root that `lpResource()` steps towards, so a model at equilibrium reproduces it exactly.

Strength of larval competition

The elasticity of somatic growth to fish abundance, at fixed spectrum shape, letting the plankton re-equilibrate:

$$\varepsilon(w) = \frac{\ln g(w; 2N, N_{pp}^*(2N)) - \ln g(w; N, N_{pp}^*(N))}{\ln 2},$$

where $N_{pp}^*(N)$ is the root above evaluated with the grazing mortality d that the fish abundance N generates. Reported as the mean over species at $w = w_1$ (egg size) and $w = 1$ g. It is zero if growth is independent of how many fish are competing, which is what “larval competition is inert” means.

Matching the fishery across variants

Contesting the plankton changes the community’s size structure so much that a common w_f in grams is not a common experiment. Instead w_f is chosen per variant so that the same *fraction of community biomass* is exposed:

$$\varphi(w) = \frac{\sum_j: w_j \geq w \sum_i w_j N_{ij} \Delta w_j}{\sum_j \sum_i w_j N_{ij} \Delta w_j}, \quad w_f^X = \arg \min_w |\varphi_X(w) - \varphi_{\text{pub}}(400 \text{ g})|,$$

with $\varphi_{\text{pub}}(400 \text{ g}) = 0.483$, giving w_f of 402 g, 10 g and 1 g for $r_0 = 10, 3, 1$.

Compensatory recruitment

`setBevertonHolt()` imposes

$$R_{dd} = \frac{R_{di} R_{\max}}{R_{di} + R_{\max}}$$

while rescaling `erepro` so that the initial state is preserved exactly. With the reproduction level $L = R_{dd}/R_{\max}$ evaluated at the initial state,

$$R_{\max} = R_{di} \frac{1-L}{L}, \quad \frac{d \ln R_{dd}}{d \ln R_{di}} = \frac{R_{\max}}{R_{di} + R_{\max}} = 1-L,$$

so $L = 0$ is the paper — recruitment tracks egg production one for one — and $L \rightarrow 1$ is recruitment pinned at $R_{\max}$ regardless of spawning stock. Preserving the steady state requires $\text{erepro}(L) = \text{erepro}(0) / (1-L)$.

Numerics

Sweeps and assembly use $\Delta t = 0.01$ yr; the final relaxation of each assembled ecosystem uses the paper’s $\Delta t = 0.002$ yr. `tests/test_convergence.R` checks that the two agree to 0.3% and that the closed-form plankton step agrees with the paper’s explicit Euler to 0.02%. Saved time steps are every 5 years. Stability is judged from an unfished 200-year projection as the coefficient of variation of total biomass over years 100–200; limit-cycle amplitude and period come from a 600-year projection, using years 300–600 and the mean spacing of successive maxima.

Phase 1 — the calibration criterion does not matter

I had called the paper's choice to match total yield *after 50 years* “the single most load-bearing arbitrary choice”. **That was wrong.**

Recalibrating against cumulative yield over the 50 years instead of terminal yield moves the constants by about 2%:

	terminal yield	cumulative yield
c_P	0.918	0.901
$c_{P/B}$	0.263	0.258

And the stronger version of the test — sweeping each rule's intensity over an 80-fold range and comparing whole frontiers rather than single points — shows the frontiers **never cross**. At matched terminal yield, the worst-affected species retains (as a fraction of its unfished trajectory):

ecosystem	fixed F	BH_P	$BH_{P/B}$
eco1	0.166	0.566	0.061
eco_r1	0.106	0.490	0.012
eco_r2	0.0031	0.345	7e-6
eco_r3	0.0104	0.297	5e-7

The shape matters as much as the ranking. Over an 80-fold increase in fishing intensity, BH_P 's worst species declines gently, from 0.85 to 0.09, while fixed F and $BH_{P/B}$ fall super-exponentially — in eco_r2, from 0.58 to 1e-18 and from 0.51 to below 1e-100. That is precisely the density dependence the paper describes: as a stock thins its own fishing mortality falls, so it cannot be driven to zero, whereas under a constant F or a constant exploitation ratio nothing arrests the decline.

So the conclusion is robust to the calibration choice, and more robustly true than the paper's single-point comparison shows. Section 3.1 of the plan is answered and closed.

One qualification. On the whole-assemblage distortion measure (RMS of $\log(B/B_{\text{control}})$), BH_P is better than fixed F in three ecosystems out of four but *worse* in eco_r1 (1.34 against 1.01). It protects the rare species by fishing the abundant ones considerably harder, which moves the community more overall. The paper reports only the rare-species outcome.

Phase 2 — the mechanism is not quite the one claimed

Four controls the paper does not run, all on eco1 at the same reference yield:

rule	worst species vs control	species below 10%	RMS log
fixed F ($\theta = 0$)	0.166	0	0.943

rule	worst species vs control	species below 10%	RMS log
BH _P	0.566	0	0.408
BH _{P/B}	0.061	2.6	1.352
$F \propto B$	0.600	0	0.400
frozen BH_P allocation	0.411	0	0.465
frozen BH _{P/B} allocation	0.049	3.3	1.426
$\theta = -0.5$	$\sim 1e-98$	5.1	15.3
$\theta = 0.5$	0.579	0	0.402
$\theta = 1.5$	0.620	0	0.412

Production is doing no work. $F \propto B_i$ performs identically to $F \propto P_i$ — marginally better, in fact. Since $P_i \sim B_i$ across species, the active ingredient is proportionality to biomass, and production is a proxy for it. This matters practically: the paper’s own discussion argues for production partly because it “automatically integrates over all the paths by which biomass flows into components of an ecosystem”, but biomass is very much easier to estimate, and in this model it does the same job.

On this ecosystem, most of the benefit looks like the allocation rather than the feedback. Freezing BH_P’s year-0 allocation and holding it constant for 50 years reaches 0.411, against 0.566 for the adaptive rule and 0.166 for fixed F — about three-quarters of the benefit on a log scale.

This does not survive replication. Across thirteen ecosystems the fraction recovered ranges from -4.66 to 0.94 with a median of 0.38 , and the frozen rule is often worse than fishing everything at the same rate. See Phase 5 below; eco1 turns out to be near the favourable end of the range, and the adaptive feedback matters much more than this single ecosystem suggests.

The benefit saturates, and only the sign of θ is critical. Writing the family as $F_i = c B_i^\theta$, the paper’s three rules are $\theta = 1$ (BH_P), $\theta = 0$ (fixed) and θ slightly negative (BH_{P/B}). Sweeping θ : -0.5 is catastrophic, 0 is the fixed- F baseline, and 0.5 , 1 and 1.5 are all much the same. A harvest control rule needs *some* positive density dependence; how much, past about $\theta = 0.5$, hardly matters.

Phase 3 item 8b — you cannot switch on larval competition and keep the paper’s ecosystem

Appendix B nominates competition among larvae for plankton as the mechanism that replaces an imposed stock-recruitment relationship. Section 3.6 of the plan showed that at the published parameter values it is inert. The question was whether switching it on changes the comparison between rules.

First, a correction to that section. Solving the steady state of Eq. (A.11) properly, at the published $r \approx 79 \text{ yr}^{-1}$ over the larval prey range the resource level is 1.004 with immigration

at its published value and 0.991 with immigration switched off entirely. $I_0 = a_0$ **is not the decisive choice I claimed**; what pins the plankton at carrying capacity is that it regenerates two orders of magnitude faster than it is grazed. Immigration only matters once r_0 is already low.

Slowing the plankton with primary production held fixed ($r_0 a_0$ constant), in a single-species screen:

r_0	egg-stage growth elasticity	long-run behaviour
10 (published)	−0.009	stable fixed point
3	−0.008	stable fixed point
1	−0.063	limit cycle, 1.33–4.63 g m ^{−2} , period ~60 yr
0.5	−0.152	limit cycle, 0.46–5.20 g m ^{−2} , period ~60 yr

There is a Hopf bifurcation between $r_0 = 3$ and $r_0 = 1$, and larval competition becomes appreciable at the same place. The cycle period exceeds the paper’s entire 50-year experiment.

Assembling full ecosystems from the same seed as eco1, so that the invader draws are identical and only the plankton differs:

	species	$w_{\max}$ range	egg elasticity	stable?
published ($r_0 = 10$)	15	327 g – 13.6 kg	−0.013	yes
$r_0 = 3$	5	112 – 161 g	−0.085	yes
$r_0 = 1$	6	112 – 359 g	−0.125	yes

Two things happen at once. Larval competition does switch on, six to ten times stronger. But **large-bodied species are excluded entirely**: with the plankton contested, juvenile growth is crushed, and no species reaches even 400 g. The top third of community biomass sits at 12 g and 1.6 g in the slowed variants, against 542 g in the published one — so the paper’s 400 g fishery would catch literally nothing.

That is a more fundamental result than the one the experiment was designed to find. **The paper’s ability to assemble a community spanning 100 g to 40 kg, and therefore to have a large-fish fishery to talk about at all, depends on the plankton being effectively inexhaustible.** The same assumption that switches off the recruitment brake is what permits large-bodied life histories.

Two smaller observations. The 60-year limit cycle seen in monoculture does not survive assembly: all three assembled communities sit at stable fixed points, so sequential assembly acts as a stability filter, which is what Appendix B argues it should. And richness falls from 15 to 5–6, with the assembly algorithm hitting its 40-attempt cap in both slowed variants.

The rule comparison, once the communities have comparable fisheries

Matching w_f on body mass is meaningless when the size structures differ this much, so it is matched on the fraction of community biomass exposed (48%, the value the published ecosystem has at 400 g), giving w_f of 402 g, 10 g and 1 g. Reference yields then come out comparable (0.198, 0.216, 0.126 g m⁻² yr⁻¹), so the three variants are being fished about equally hard.

variant	rule	worst species	RMS log
published ($r_0 = 10$)	fixed F	0.166	0.943
	BH_P	0.566	0.408
	BH _{P/B}	0.061	1.352
$r_0 = 3$	fixed F	0.765	0.150
	BH_P	0.940	1.029
	BH _{P/B}	0.727	0.189
$r_0 = 1$	fixed F	0.969	0.033
	BH _P	0.604	1.947
	BH _{P/B}	0.892	0.121

Two things change, and the second is the headline.

Fishing stops mattering much at all. With the plankton contested, the worst-affected species retains 0.6 to 0.97 of its unfished biomass under every rule, against 0.06 to 0.57 in the published ecosystem. A real density-dependent brake protects species from fishing regardless of how the fishing is distributed, which is exactly what section 3.2 of the plan predicted would happen if compensatory recruitment were restored by any route.

The ranking inverts. At $r_0 = 1$, BH_P leaves its worst species at 0.604 while fixed F leaves it at 0.969, and BH_P distorts the assemblage roughly sixty times more on the RMS-log measure (1.95 against 0.03). The mechanism is not mysterious: $F \propto B$ concentrates fishing on the abundant species, and when every species already has its own brake, that concentration is pure distortion with nothing to buy. BH_P's advantage exists because the model has no other density dependence for it to be redundant with.

How much weight this carries

Less than the numbers alone suggest, and the reason is structural rather than statistical. Contesting the plankton does two things at once: it restores a recruitment brake, *and* it replaces a 15-species community spanning 327 g to 13.6 kg with a 5- to 6-species community of fish under 360 g. These cannot be separated within this experiment, because as shown above there is no parameter setting that gives one without the other. So this is evidence that the conclusion is contingent on the absence of a recruitment brake, not proof of it.

The clean test is Phase 3 item 8: impose a Beverton-Holt stock-recruitment relationship on the *published* ecosystem, where the community is held fixed and only the density dependence changes. That is the experiment to run next, and this result raises its priority considerably.

Phase 3 item 8 — the result is contingent on the absence of compensatory recruitment

This is the experiment item 8b could not do. `setBevertonHolt()` imposes a stock-recruitment relationship on the *published* ecosystem while rescaling `erepro` so that the initial state is preserved exactly — verified to 2×10^{-4} — so the community, its 15 species, its size structure and its steady state are **identical across the whole sweep**. Only the sensitivity of recruitment to egg production changes, from $d \ln R_{dd} / d \ln R_{di} = 1$ at $L = 0$ (the paper) to 0.1 at $L = 0.9$.

Worst-affected species, as a fraction of its unfished trajectory, at matched terminal yield:

L	recruitment elasticity $1 - L$	fixed F	BH _P	BH _{P/B}	BH _P advantage
0 (the paper)	1.00	0.166	0.566	0.061	3.41×
0.25	0.75	0.337	0.703	0.170	2.09×
0.50	0.50	0.603	0.777	0.395	1.29×
0.75	0.25	0.778	0.824	0.635	1.06×
0.90	0.10	0.834	0.846	0.730	1.01×

The advantage closes monotonically and essentially vanishes. At the paper's $L = 0$, BH_P leaves its worst species 3.4 times better off than fixed F does. By $L = 0.5$ — where halving egg production still costs 29% of recruitment, which is mild compensation by the standards of fitted stock-recruitment relationships — the factor is 1.29. By $L = 0.9$ it is 1.01. The same happens to the whole-assemblage measure: the RMS-log ratio between fixed F and BH_P falls from 2.31 to 1.42. And the species that BH_{P/B} drives below 10% of control at $L = 0$ — 2.6 of them — are all rescued by $L = 0.25$ alone, with no change to the harvest rule at all.

So the density dependence that BH_P supplies through the harvest rule is largely redundant with the density dependence that a stock-recruitment relationship supplies through the population. The paper's model has none of the latter, by deliberate design, and that is why the former looks so valuable.

This also reconciles item 8b. There the ranking *inverted*; here it does not — BH_P stays nominally best at every level, just by a shrinking margin. The inversion in item 8b is therefore attributable to the community change (a 5-species assemblage of fish under 360 g) rather than to the recruitment brake itself. Two experiments that looked like they

disagreed actually separate cleanly: the brake shrinks the advantage, the community change reverses it.

One caveat on the far end of the sweep. Preserving the steady state requires $\epsilon_{\text{repro}}(L) = \epsilon_{\text{repro}}(0) / (1 - L)$, so $L = 0.9$ needs $\epsilon_{\text{repro}} = 4.0$, i.e. a reproductive efficiency $\epsilon_R = 2.0 > 1$ — more egg mass than the energy budget allows. That row is a limiting case, not a plausible parameterisation. The physically comfortable range is $L \leq 0.5$ ($\epsilon_R \leq 0.4$), and the advantage is already down to $1.29\times$ there.

A side effect worth recording: compensation also damps the model’s own transients. Maximum unfished drift over 50 years falls from 83% at $L = 0$ to 21% at $L = 0.9$, which is further evidence that the quasi-equilibrium the paper starts from is as unsettled as it is partly *because* recruitment is uncompensated.

Phase 5 — replication across thirteen ecosystems

Everything above rests on one to four assembled ecosystems. Twelve further ecosystems were assembled under exactly the eco1 protocol — no randomisation of the search rate, $z'_i = 1$, $F_{\text{ref}} = 0.1$ — giving thirteen comparable replicates, and the frontier comparison was repeated on all of them for the paper’s three rules plus the two Phase 2 controls that carried the load.

All thirteen reached the full 15 species within the 40-attempt cap (18–30 attempts, median 25, against the paper’s 25), each spanning roughly two decades of maximum body mass, with 14 or 15 species above the 400 g entry size. No assemblage had to be screened out, so the distributions below are over every replicate rather than a chosen subset.

Worst-affected species, as a fraction of its unfished trajectory, read at each ecosystem’s own reference yield:

rule	median	range
fixed F	0.234	0.124 – 0.460
BH_P	0.593	0.380 – 0.670
BH _{P/B}	0.081	0.036 – 0.359
$F \propto B$	0.660	0.522 – 0.717
frozen BH _P allocation	0.372	0.009 – 0.585

What replication confirms

The ranking is completely robust. BH_P beats fixed F in 13 of 13, beats BH_{P/B} in 13 of 13, and the full ordering BH_P > fixed > BH_{P/B} holds in 13 of 13. The paper’s qualitative conclusion survives replication without exception.

The effect size varies fourfold. BH_P’s advantage over fixed F in the worst-affected species has a median of **2.42×**, ranging from $1.27\times$ to $5.24\times$. eco1, the ecosystem all the

earlier phases were reported on, gives $3.41 \times$ — comfortably above the median. Single-ecosystem effect sizes should be read as illustrative, not as estimates.

Production really is doing no work. $F \propto B_i$ beats $F \propto P_i$ in **13 of 13** ecosystems, by a median of 10% and never by less than 6%. Phase 2 called this a tie on one ecosystem; with thirteen it is a consistent, if small, win for the simpler and far more measurable quantity.

One reading of that would be an artefact: across these thirteen ecosystems P/B varies by less than a factor of two, so $F \propto P$ and $F \propto B$ are nearly the same rule and the gap between them could be noise. [life-history.md](#) tests the reading on assemblages built so that P/B varies by 5 to 8x, where the two allocations differ by a measured factor of z . $F \propto B$ still wins, and by more on the rarest species. The finding is not an artefact of flat P/B .

What replication overturns

The frozen-allocation result does not hold. On eco1, freezing BH_P 's year-0 allocation recovered 74% of the gap between fixed F and adaptive BH_P , which Phase 2 read as “most of the benefit is the allocation, not the feedback”. Across thirteen ecosystems the fraction recovered has a median of 0.38 and a range of **−4.66 to 0.94**. A negative value means the frozen allocation is *worse than fishing every species at the same rate*, and that happens in 4 of 12 readable cases; frozen beats fixed F in only 8 of 12. In eco303 the frozen rule cannot reach the reference yield **at any intensity** — pushing harder collapses the stocks and total yield falls — so there is no intensity at which it is comparable at all.

eco1's 0.74 sits near the top of that range. Generalising from it was wrong.

The mechanism is clear enough in hindsight. A frozen allocation assigns high fishing mortality to whatever was abundant in year 0 and never revises it, so a species that starts abundant and then declines keeps being hit at its original rate — the same failure mode as fixed F , but concentrated on the species the rule singled out. The adaptive feedback is not a refinement on top of a good allocation; it is what stops the allocation going stale. **This vindicates the paper's emphasis on adaptation**, which Phase 2 had wrongly downplayed.

A methodological correction

Terminal yield is not always monotone in fishing intensity. Rules without a stabilising feedback overfish into declining yield, so their frontiers double back: 6 of 65 frontiers here peak at an intermediate intensity, and they are all $BH_{P/B}$ (3) or frozen (3). BH_P , $F \propto B$ and fixed F are monotone in all thirteen — itself a result worth stating, since it means the density-dependent rules do not have an interior yield maximum to overshoot over this range.

Interpolating a metric against yield across such a turnover mixes the two branches.

`lp_at_yield()` now restricts to the ascending branch up to the yield maximum, and returns `NA` — meaning “unreachable at any intensity” — rather than extrapolating. This changed $BH_{P/B}$'s reported range in this section from a spurious 7×10^{-8} low end to 0.036, and moved the frozen-versus-fixed count from 6/12 to 8/12. Re-deriving Phases 1 and 2 with the

corrected reader leaves every number in them unchanged, because their reference yields sit on the ascending branch in all cases.

Phase 4 — the rule tolerates infrequent updating, but not bad estimates

Phase 5 established that the adaptive feedback, not the initial allocation, is what makes BH_P work. That makes it matter how well the feedback survives being applied the way a real fishery would apply it: recalculated every few years from survey estimates, rather than continuously from perfect knowledge. Two degradations were imposed together on five ecosystems — an update interval, and mean-preserving log-normal error of log-scale s.d. σ redrawn independently at every update.

BH_P's advantage over fixed F in the worst-affected species:

update interval	σ	advantage, median [range]
continuous	0	1.78× [1.27, 3.41]
5 yr	0	1.78× [1.27, 3.40]
10 yr	0	1.78× [1.29, 3.39]
25 yr	0	2.06× [1.51, 3.45]
never (set once at year 0)	0	1.35× [0.92, 2.47]
5 yr	0.5	1.81× [0.66, 3.14]
10 yr	0.5	1.76× [0.66, 2.11]
5 yr	1.0	0.73× [0.00, 1.44]

Updating frequency barely matters. Recalculating F_i every 5 or 10 years is indistinguishable from recalculating it continuously — the worst-species outcome on eco1 moves from 0.566 to 0.564 to 0.563. Even a 25-year interval is fine. Only never revising the allocation degrades the rule, to 1.35× and as low as 0.92× — worse than fishing everything at the same rate.

This is a genuinely encouraging result for the paper's policy claim, and the reason is that the ecosystem's own timescale is long: the fished trajectories in Fig. 3 take decades to develop, so a ten-year-old estimate of P_i is still nearly right. A decadal survey cycle is enough to keep the feedback alive. What fails is not slow updating but *no* updating.

Estimate quality matters much more. At $\sigma = 0.5$ — a typical estimate wrong by a factor of about 1.65 — the median advantage is untouched, but the range now reaches 0.66, meaning that in one ecosystem out of five BH_P ends up *worse* than uniform fishing. At $\sigma = 1.0$, a factor of about 2.7, the median advantage falls to **0.73×**: on balance worse than fixed F , and in one ecosystem the worst species is driven to 4×10^{-4} of its unfished trajectory.

The asymmetry between the two is worth stating plainly, because it is the opposite of what the paper’s framing suggests. Law & Plank present the adaptive feedback as protection “in the presence of uncertainty about how marine ecosystems work”. It is protection against *structural* uncertainty — not knowing how the system will respond — because the rule simply follows whatever the state turns out to be. It is not protection against *observational* uncertainty: a species whose production is over-estimated is fished proportionally harder, with nothing in the rule to arrest it, so estimation error feeds straight through into mis-allocated mortality. Fixed F has no such exposure, because it never consults an estimate at all.

Whether $\sigma = 0.5$ or 1.0 is the realistic figure is an empirical question this model cannot answer, but the paper’s own discussion concedes that “the rare species of special importance for conservation are also the species for which information on biomass, production rate and fishing mortality is most likely to be scarce” — and it is precisely the rare species that the rule has to get right.

Phase 3 items 10 and 11 — the fishery’s design

Both of these change the fishery without touching the ecosystem, so the same five assembled communities are used throughout.

Item 10: the entry size w_f

The paper fishes every species from 400 g upwards and flags the single shared knife edge as a simplification. Combined with a fixed $w_{\text{mat}} = w_{\text{max}}/10$ it means a 1 kg species is fished only as an adult while a 40 kg species is fished for a decade before it breeds, so some of “large species are vulnerable” may be built into the design rather than discovered.

w_f	reference yield	fixed F	BHp	BHp/B	advantage	readable
100 g	0.525	0.542	0.647	0.356	1.18× [1.06, 1.27]	5/5
200 g	0.476	0.478	0.691	0.272	1.37× [1.17, 2.50]	5/5
400 g (published)	0.354	0.298	0.635	0.162	1.92× [1.44, 3.41]	4/5
800 g	0.204	0.206	0.686	0.092	3.71× [1.81, 3.77]	3/5

The conclusion holds at every entry size, but its size depends strongly on w_f . BHp beats fixed F in every ecosystem at every entry size tested. But the advantage grows from 1.18× at 100 g to 3.71× at 800 g — roughly threefold across the range, with the paper’s 400 g sitting in the middle.

The reason is visible in the columns: BHp’s own outcome barely moves (0.647, 0.691, 0.635, 0.686) while **fixed F degrades sharply** (0.542 down to 0.206). Raising w_f concentrates

fishing on the species with the longest juvenile phase relative to the entry size, which is exactly where a constant F causes recruitment overfishing — and exactly what BH_P 's feedback arrests, since those species' F falls as they decline. So the choice of entry size is not neutral: it sets how much damage there is for the feedback to prevent. At a low entry size, where nearly everything is fished across most of its life, balanced harvesting has little left to offer.

(The 800 g row rests on three ecosystems and the 400 g row on four: with only three intensities sampled here, the reference yield is not always bracketed. Where it is not, the rule genuinely could not reach that yield within the sampled range.)

Item 11: the range over which P and B are measured

The paper measures P_i and B_i over the harvested range only, on the grounds that “reliable information is most likely to be available over this range”. That makes P_i exclude juvenile production — most of a species' somatic production — and, for a species whose $w_{\max}$ is not far above w_f , makes it dominated by the boundary influx term at w_f . Here the measurement range is instead the whole life cycle, with the harvested range left at 400 g, so only the rule's *information* changes and not what is caught.

P, B measured over	fixed F	BH_P	$BH_{P/B}$	advantage
harvested range (published)	0.298	0.635	0.162	1.92× [1.44, 3.41]
whole life cycle	0.298	0.748	0.487	2.40× [1.63, 4.96]

Both balanced-harvesting rules do better on whole-life-cycle information, and $BH_{P/B}$ dramatically so — its worst-affected species goes from 0.162 to 0.487, a threefold improvement, purely from measuring the same quantities over a wider range.

That is worth dwelling on, because $BH_{P/B}$ is the rule the paper argues against. Part of its poor showing comes not from the idea of a constant exploitation ratio but from the restricted measurement window: P_i/B_i computed over $[w_f, w_{\max, i}]$ is nearly the same for every species, which is precisely why the paper finds it barely distinguishable from a constant F . Computed over the whole life cycle it varies much more between species — small, fast-turnover species have genuinely higher P/B — so the rule becomes discriminating and much less harmful. BH_P still wins, but the gap narrows from 3.9× to 1.5× on the worst-species measure.

The paper's data-availability argument for the restricted range is reasonable as far as it goes, but this suggests the restriction is not cost-free: it degrades both rules, and it exaggerates the difference between them.

Still to do

Phase 3 items 8 (compensatory recruitment), 9 (Eq. A.7), 10–11 (w_f and the P/B measurement range), 12–14 (θ matrix, plankton cap, grid); Phase 4 (update interval and observation error); Phase 5 (replication across 10–20 assemblages). Item 8 remains the most likely single result-changer.

Page <https://gustavdelius.github.io/balanced-harvest/robustness-results.html> of the site accompanying the mizer reimplementation of Law & Plank (2023). Generated by `make_pdfs.py`.